Week 13

MICROECONOMIC THEORY
ECON 323 502/503

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Review

A **dominant strategy** is a strategy that is optimal no matter what an opponent does.

Equilibrium in dominant strategies is the outcome of a game in which each player plays a dominant strategy.

An outcome x **Pareto dominates** another outcome y, if x makes at least one player better off without hurting any other player compared to y.

Prisoner's dilemma is a game with the following two features :
(1) it has an equilibrium in dominant strategies;
(2) the equilibrium in dominant strategies is Pareto dominated by another feasible outcome of the game.

A **Nash Equilibrium** is a profile of strategies in which each player is playing a **best response** to the other player's strategy.

Nash Equilibrium May Not Be Unique

- In the Battle of sexes example;

		Boy	
		Movie	Football
Girl	Movie	3, 2	0, 0
	Football	0, 0	2, 3

- In this game, (Movie, Movie) is a Nash equilibrium, because
  - (1) given Boy plays Movie, it is a best response for Girl to play Movie,
  - (2) given Girl plays Movie, it is a best response for Boy to play Movie.
- (Movie, Football) is not Nash equilibrium.
- (Football, Movie) is not Nash equilibrium.
- (Football, Football) is also a Nash equilibrium.

Nash Equilibrium May Not Be Unique

two Nash equilibria,
though no equilibrium in dominant strategies

- In the Battle of sexes example;

		Boy	
		Movie	Football
Girl	Movie	3, 2	0, 0
	Football	0, 0	2, 3

- In this game, (Movie, Movie) is a Nash equilibrium, because
  - (1) given Boy plays Movie, it is a best response for Girl to play Movie,
  - (2) given Girl plays Movie, it is a best response for Boy to play Movie.
- (Movie, Football) is not Nash equilibrium.
- (Football, Movie) is not Nash equilibrium.
- (Football, Football) is also a Nash equilibrium.

		Boy	
		Movie	Football
Girl	Movie	3, 2	0, 0
	Football	0, 0	2, 3

A quick way to identify all Nash equilibria (Should be very useful):

Assume the Column Player plays a strategy, put a line under the Row Player's (all) best response payoff(s).

Change the strategy of the Column Player, repeat.

Assume the Row Player plays a strategy, put a line under the Column Player's (all) best response payoff(s).

Change the strategy of the Row Player, repeat.

The outcomes with two underlines are Nash equilibria.

Practice the Quick Method to Identify <u>All Nash Equilibria</u>

		Firm 2		
		$Q = 96$	$Q = 64$	$Q = 48$
Firm 1	$Q = 96$	$\$0, \$0$	$\$3.1, \$2.0$	$\$4.6, \$2.3$
	$Q = 64$	$\$2.0, \$3.1$	$\$4.1, \$4.1$	$\$5.1, \$3.8$
	$Q = 48$	$\$2.3, \$4.6$	$\$3.8, \$5.1$	$\$4.6, \$4.6$

Practice the Quick Method to Identify <u>All Nash Equilibria</u>

		Firm 2		
		$Q = 96$	$Q = 64$	$Q = 48$
Firm 1	$Q = 96$	\$0, \$0	\$3.1, \$2.0	\$4.6, \$2.3
	$Q = 64$	\$2.0, \$3.1	\$4.1, \$4.1	\$5.1, \$3.8
	$Q = 48$	\$2.3, \$4.6	\$3.8, \$5.1	\$4.6, \$4.6

- Step 1: Underline Player 1's (all) best-response payoffs when Player 2 plays Strategy 96, 64, and 48 respectively.
- Step 2: Underline Player 2's (all) best-response payoffs when Player 1 plays Strategy 96, 64, and 48 respectively.
- Step 3: Every vector with two underlines is a Nash equilibrium.

Practice the Quick Method to Identify <u>All Nash Equilibria</u>

		Firm 2		
		$Q = 96$	$Q = 64$	$Q = 48$
Firm 1	$Q = 96$	\$0, \$0	\$3.1, \$2.0	\$4.6, \$2.3
	$Q = 64$	\$2.0, \$3.1	\$4.1, \$4.1	\$5.1, \$3.8
	$Q = 48$	\$2.3, \$4.6	\$3.8, \$5.1	\$4.6, \$4.6

- Step 1: Underline Player 1's (all) best-response payoffs when Player 2 plays Strategy 96, 64, and 48 respectively.
- Step 2: Underline Player 2's (all) best-response payoffs when Player 1 plays Strategy 96, 64, and 48 respectively.
- Step 3: Every vector with two underlines is a Nash equilibrium.

Nash Equilibrium May Not Exist (We do not consider randomization)

		Evan	
		Movie	Gym
Kathy	Movie	-1, 1	1, -1
	Gym	1, -1	-1, 1

None of the four outcomes is Nash equilibrium.

Equilibrium in Dominant Strategies vs Nash Equilibrium

Equilibrium in Dominant Strategies: I'm doing the best I can no matter what you do. You're doing the best you can no matter what I do.

Nash Equilibrium: I'm doing the best I can given what you are doing. You're doing the best you can given what I am doing.

An equilibrium in dominant strategies is always a Nash equilibrium, but not the other way around.

Maximin Strategies

		Firm 2	
		DON'T INVEST	**INVEST**
Firm 1	DON'T INVEST	0, 0	-10, 10
	INVEST	-100, 0	20, 10

(Invest, Invest) is the only Nash equilibrium of this game.

Nash equilibrium requires rationality of all players. But if Firm 1 is afraid that Firm 2 makes a mistake and chooses "Don't invest", Firm 1's payoff from "Invest" will be extremely low.

Afraid of the worst-case scenario, Firm 1 may take a conservative strategy, which is "Don't invest". This motivates the maximin strategy.

Maximin Strategies

- **Maximin Strategy** Strategy that maximizes the minimum gain that can be earned.
- **An equilibrium in maximin strategies** is a profile of strategies in which all players play a maximin strategy.

Maximin Strategies

		Firm 2	
		DON'T INVEST	**INVEST**
Firm 1	DON'T INVEST	0, 0	-10, 10
	INVEST	-100, 0	20, 10

- Firm 1's maximin strategy:
  - By Don't Invest, Firm 1 may get 0 or -10. Worst case is -10.
  - By Invest, Firm 1 may get -100 or 20. Worst case is -100.
- Don't invest is the maximin strategy for Firm 1.

- Firm 2's maximin strategy:
  - By Don't Invest, Firm 2 gets 0 or 0. Worst case is 0.
  - By Invest, Firm 2 gets 10 or 10. Worst case is 10.
- Invest is the maximin strategy for Firm 2.

- (Don't Invest, Invest) is the equilibrium in maximin strategies

Maximin Strategies

		PLAYER B	
		WORK	SHIRK
PLAYER A	WORK	8, 8	8, 2
	SHIRK	2, 8	10,10

For Player A,

- by Working, Player A's worst-case payoff is 8;
- by Shirking, his worst-case payoff is 2.
- 8>2, so Working is Player A's maximin strategy.

For Player B,

- by Working, Player B's worst-case payoff is 8
- By Shirking, his worst-case payoff is 2
- 8>2, so Working is a maximin strategy for Player B.

Hence, (Work, Work) is an equilibrium in maximin strategies.
(Work, Work) and (Shirk, Shirk) are both Nash equilibria.

Practice Problem 1

		PLAYER 2	
		L	R
PLAYER 1	U	0, 5	2, -10
	D	1, -5	4, 4

Does this game have an equilibrium in dominant strategies?

A. Yes
B. No

Answer: B

Player 1 has a dominant strategy D, but player 2 does not have a dominant strategy.

Practice Problem 2

		PLAYER 2	
		L	R
PLAYER 1	U	0, 5	2, -10
	D	1, -5	4, 4

How many Nash equilibria are there in the game?

A. 0
B. 1
C. 2
D. 3

Answer: B

It is easy to verify that only (D, R) is the Nash equilibrium.

Practice Problem 2

		PLAYER 2	
		L	R
PLAYER 1	U	0, 5	2, -10
	D	1, -5	4, 4

What is the maximin strategy of Player 2?

A. L

B. R

Answer: A.

By playing L, Player 2's worst-case payoff is -5; by playing R, Player 2's worst-case payoff is -10. L is a maximin strategy of Player 2 because it maximizes Player 2's worst-case payoff.